

THE SOLUTION – SUBSYSTEM MOATS & BENCHMARK

AI Swarm Drones Find Survivors 3–4x Faster Than Traditional SAR

- 78-85% survivor detection on first pass
- 10-18 min per sq mile speed
- Mesh networking works in GPS-denied zones
- 24/7 autonomous operation for 11+ days
- False positive rate down to 12%
- Cost per deployment: \$12K vs \$15K

PROJECT SUTRA SUBSYSTEM
SOLUTIONAL MOATS

SUBSYSTEM
A MOAT

PX4 +
VIO
EKF2

SUBSYSTEM
B MOAT

Deep
JSCC
Video

3 CORE
MOATS

SUBSYSTEM
C/D MOAT

3D DEM
Raycast

SWARM SEARCH
EFFICIENCY

BENCHMARK MATRIX

Search & Rescue Performance Comparison

EMPIRICAL
BENCHMARK

METRIC	TRADITIONAL DRONE	AI SWARM (SUTRA)
Deployment Time	45-90 min	15-30 min
Area Coverage / mi ²	2-3 hours	45-60 min
Detection Rate	65-70%	55-65%
Personnel Required	15-25 crew	2-3 pilots

8-15
min

10-18
min

78-85%

1-2
operators

SPEED & EFFICIENCY COMPARISON

3-4X FASTER
COVERAGE

Area Survey Rate 4.5 mi²/hr vs
(sq miles / hr) 0.4 mi²/hr

Traditional Ground 0.4 mi²

Single Drone 1.2 mi²

AI Swarm (SUTRA) 4.5 mi²

\$8.94B
Market by
2033

37
Nations
Mandated
SAR Drones

\$180M
FEMA 2025
Budget

50Hz
Offboard
GNC, GPS-
Denied
Flocking

Zero-
Cliff
Video
Down to
-5dB SNR

3.59cm
Elevation-
Corrected
Geolocation